# Alice Petrosyan

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## Technical Skills

**Design**: SolidWorks, Solid Edge, OnShape, StarCCM+, AutoCAD, DraftSight, MicroStation, GD&T
**Programming**: Python, C/C++, PLC, VHDL, ReactJS, HTML/CSS, VisualBasic

## Experience

### Controls Engineering Intern
January 2026 – April 2026

*Tesla* — *Markham, ON*

- Implemented, debugged, and optimized PLC control logic using **Siemens TIA Portal**, integrating with safety devices, servo drives, and HMI interfaces, to support machine sequencing and safety interlocks
- Developed HMI interfaces for 10+ production stations to monitor machine performance, access diagnostics, and control production equipment
- Automated HMI deployment processes using **PowerShell** and **FTP**, enabling simultaneous deployment of multiple HMI instances, reducing download times by 200%

### Controls Engineering Intern
May 2025 – August 2025

*Apex Motion Control* — *Surrey, BC*

- Developed **Allen Bradley PLC** programs and **HMI**s to deposit a variety of products, improving consistency of deposit volumes to an accuracy of 98%
- Engineered **UR** robot programs using **Python**, **XML-RPC**, and **MODBUS**, reducing camera calibration times by 75%
- Drafted wiring diagrams for 10+ machines using **DraftSight**, facilitating cross-team communication and streamlining the manufacturing process

### Robotics Design Intern
September 2024 – December 2024

*Apex Motion Control* — *Surrey, BC*

- Implemented **UR** and **Doosan** robot scripts to generate cake decorating paths of variable shape and size, to automate the frosting of cakes along a conveyor belt, increasing production rates by 300%
- Developed path visualizer using **Matplotlib**, to calculate angle of end-of-arm-tooling along cake decorating path
- Designed and tested **end-of-arm-tooling** using **SolidWorks**, to evenly distribute buttercream onto cakes of various shapes and sizes, improving cake decorating consistency and frosting distribution
- Programmed stack lights using **Revolution Pi** and **ladder logic** to indicate current status of machines, allowing operators to quickly check robot status

### CAD Technologist
January 2024 – April 2024

*S&C Electric* — *Etobicoke, ON*

- Developed and implemented detailed standard work instructions in **PowerPoint** and **Excel**, collaborating closely with assemblers to enhance quality control and standardize assembly line processes
- Designed, built, and installed workstations using **Solid Edge** to provide assemblers with a more intuitive, organized, and minimalistic workspace
- Created macros using **Visual Basic** to automatically retrieve data and update status on current job orders and missing parts, increasing tracking efficiency

### Aerodynamics Team Member
September 2023 – Present

*University of Waterloo Formula Electric* — *Waterloo, ON*

- Designing and validating 2026 rear wing using **SolidWorks** and **StarCCM+** to increase downforce and reduce drag
- Lead composites manufacturing sessions to produce airfoils and flat panels for the aero package
- Collaborating with other subteams to integrate aerodynamic solutions with overall chassis design, ensuring optimal balance between downforce generation and vehicle handling characteristics

## Education

**University of Waterloo** — Waterloo, ON

**BASc Honours Mechatronics Engineering**
**Relevant Coursework:** DSA, RTOS, Sensors & Instrumentation, Power Electronics, Microprocessor Systems and Interfacing, MODS, Thermodynamics & Heat Transfer, Systems & Signals
**Awards:** Dean's List, President's Scholarship of Distinction, President's Research Award, Governor General's Bronze Medal